Real-Time Retail Feedback Intelligence

MIT Capstone Project Presentation



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CONTENTS

Presentation Roadmap

MAIN PRESENTATION

- Executive Summary 03
- **01 | Business Problem → AI Solution** 04
- **02 | Evidence Base** 05
- **03 | Controlled Evaluation** 06
- **04 | Solution Design** 07
- **05 | Evaluation** 08

CONTINUED / APPENDIX

- **06 | Results & Cost Analysis** 09
- **07 | Business Value & Recommendation** 10
- **08 | Final Outcome & Recommendation** 11

APPENDIX

- Appendix 01 | How Macro-F1 Is Calculated 12
- Appendix 02 | How the Judge Score Is Derived 13

Executive Summary

Problem Statement

- ChicStyle, a growing fashion retail platform, faces huge spikes in customer reviews during festive seasons
- Reviews include praise and urgent issues (fit, sizing, defects, delivery delays) that require timely responses.
- ChicStyle needs a system to process large review volumes quickly with business context.

Implementation Summary

- Built an end-to-end GenAI pipeline to convert review text into sentiment, issue category, summary, urgency, personalized response, and retail insight.
- Engineered and evaluated multiple prompting techniques using qualitative and quantitative methods.
- Used a 50-customer review benchmark to compare seven prompting strategies.
- Re-validated the strongest approaches on 180 customer reviews using a controlled stratified design.

Outcome and Recommendation

- COT2 -Chain of Thought is recommended
- For production deployments **validation gates are required**

Assumptions, Limitations and Weaknesses

- AI-assisted sentiment labels may involve interpretation and subjectivity.
- LLM-as-a-Judge provides an additional quality perspective; it does not replace human evaluation.
- Business recommendations are directional, not quantified estimates of sales, returns, or financial impact.
- Production deployment requires larger validation, monitoring, governance, cost controls, and human oversight.
- Quality weights (55% Macro-F1, 20% Judge, 15% nuance, 10% business usefulness) are task-specific, not universal standards.
- Next step: Conduct sensitivity analysis to test if COT2 remains preferred under different weighting schemes.

From Customer Feedback to Business Action

This project turns customer reviews into actionable feedback through an automated GenAI workflow.

1	**Understand the data**	22,641 usable reviews from the '*Women's E-Commerce Clothing Reviews*' dataset were analyzed to understand sentiment patterns.
2	**Experiment with prompts**	Engineer and evaluate seven prompting strategies on 50 benchmark customer reviews.
3	**Validate the finalists**	Compare FS2 and COT2 on a larger 180-review set, balanced across sentiment and review length.
4	**Separate key signals**	Sentiment describes the experience; recommendation describes the purchase decision.
5	**Convert insight into action**	Use urgency, issue themes, and summaries to guide service responses and product improvements.

Goal: Reduce the time from *customer feedback → understanding → action → improvement*.

Key Insights from Customer Reviews

23,486
original records

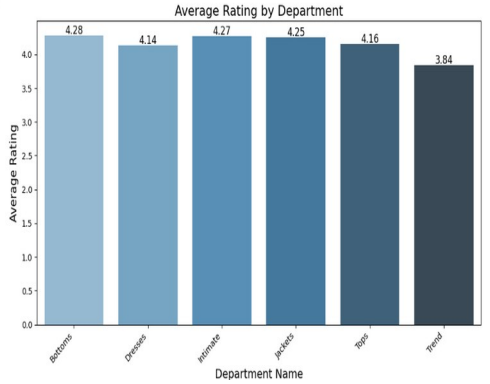
845
records with missing review text removed

22,641
usable review texts

3
sentiment classes (Positive, Negative, Neutral)

Exploratory data analysis - Insights

- Review.Text is the primary field analyzed for customer sentiment.
- Ratings are strongly positive, creating an imbalanced rating-based signal.
- Tops and Dresses receive the most feedback.
- Trend has the lowest department average rating: 3.84.
- Low-rated reviews frequently mention sizing, fit, material quality and mismatched images.



Department average rating



Language patterns in ratings

Takeaways:

- Ratings are strongly skewed toward positive feedback, so controlled evaluation uses balanced representation across sentiment classes.
- Rating and written-review sentiment capture different signals and should not be treated as interchangeable.

Controlled Evaluation of Prompt Designs

Prompts were developed and refined → tested on 50 benchmark reviews → top two validated on 180 reviews.

Prompt Designs

ZERO-SHOT

ZS1 • ZS2

Instructions only. No examples.

FEW-SHOT

FS1 • FS2

Instructions with labeled examples

CHAIN-OF-THOUGHT

COT1 • COT2

Prompted to show reasoning. No examples.

HYBRID

HYB

Structured reasoning with labeled examples.

STAGE 1 — 50-REVIEW BENCHMARK

- Fixed set of 50 labeled benchmark customer reviews.
- Balanced sentiment: 16 reviews per sentiment + 2 additional Positive; Short/Medium/Long mix preserved within each sentiment.
- All seven prompt strategies evaluated on the same reviews and the same six output fields.
- Written-review sentiment is the primary criterion; rating alignment and Judge scores provide supporting evidence.

STAGE 2 — 180-REVIEW REVALIDATION

- Fixed set of 180 labeled benchmark customer reviews.
- Equal representation across 9 Sentiment × Review-Length* strata — 20 reviews per stratum.
- Final evaluation compares the two strongest prompt strategies: FS2 and COT2.

*Review length: Short (0–25 words) | Medium (26–60 words) | Long (>60 words)

Within each controlled evaluation, model, data, output, and evaluation settings were held constant so observed performance differences primarily reflect prompt design.

From One Review to Structured Customer Intelligence

CUSTOMER REVIEW

Example : “Beautiful fabric and flattering design, but the size is much too large. I will return it.”

GenAI Model

The model must interpret the overall experience — not isolated positive or negative phrases.

Structured output

1 **Sentiment**

Applicable: Positive / Negative / Neutral
Example: Negative

2 **Category**

Applicable: Type of feedback expressed in the review
Example: Fit / Sizing

3 **Feedback Summary**

Applicable: Concise customer-level summary
Example: Fabric and design praised; size is too large and will be returned.

4 **Urgency**

Applicable: High / Medium / Low
Example: Medium

5 **Personalized Message**

Applicable: Customer-facing response
Example: Sorry the fit did not meet your expectations.

6 **Retail Team Insight**

Applicable: Actionable business insight
Example: Review sizing guidance for this product.

Model Evaluation Criteria

PRIMARY Evaluation

- 50 and 180 benchmark reviews -AI-assisted sentiment labeling
- Model predictions compared with the reference labels using Macro-F1
- Macro-F1 is the primary measure of sentiment performance

SUPPORTING — LLM JUDGE Evaluation

- Judge sees original customer review + model output
- Scores correctness, nuance, grounding, completeness and usefulness
- Provides an additional quality perspective beyond sentiment scoring

DIAGNOSTIC — RATING ALIGNMENT Evaluation

- Compares model sentiment with rating-derived sentiment
- Useful for consistency
- Not text-sentiment truth: rating and written experience can differ

Guardrails: “Mixed” is evidence within a review, not a fourth sentiment class.
Written-review sentiment is primary; rating alignment is secondary.

The weights and the dimensions used for evaluations are explained in Appendix

What the Larger Run Changed

Initial benchmark: FS2 was strongest. Larger, evenly stratified revalidation: COT2 was more consistent.

Strategy	180 Accuracy	180 Macro-F1	180 Final Score	Gen cost / 1K	Avg latency
COT2	90.29%	79.73%	**62.46%**	**$0.125**	**1.20 s**
FS2	87.22%	**75.42%**	**59.47%**	**$0.168**	1.06 s

INITIAL 50-REVIEW BENCHMARK

FS2: 90.0% accuracy
86.52% Manual Text Macro-F1
88.12% final score

COT2: 86.0% accuracy
80.77% Manual Text Macro-F1
85.00% final score

WHAT CHANGED

COT2 leads the larger run on text-sentiment quality, composite quality and rating alignment, while using fewer tokens and costing less to generate.

Trade-off: FS2 is slightly faster; COT2 provides the stronger quality/cost balance in the current 180 run.

Classifier and Business Recommendation GenAI Models

Recommendation-Based GenAI I (0/1 binary classifier)

• Uses the customer review text to predict **Recommended (1) or Not Recommended (0)**
• A separate GenAI prompt was designed specifically for this binary task
• The model returns a **0/1 recommendation + brief reason**
• The actual recommendation label is hidden from the model to avoid data leakage
• The model recommendation is evaluated against the actual recommendation

GENAI BINARY CLASSIFIER EVALUATION

Result: 88.0% Accuracy | 90.9% Macro Precision | 87.0% Macro Recall | 87.5% Macro-F1

The classifier was conservative: it made no false-positive recommendation predictions. The Model missed 6 reviews that were actually marked Recommended. Recommendation ≠ Sentiment: they capture different aspects of customer feedback.

Confusion Matrix: Shows where predictions were correct and where the model confused 0 and 1. **Accuracy:** Overall percentage of correct predictions.
Macro Precision: Average precision across the two recommendation classes.
Macro Recall: Average recall across the two recommendation classes.
F1: Balances precision and recall.

GenAI - Business Recommendations for COT2

Short term | 3–6 months

- Strengthen sizing & fit guidance
- Improve care instructions and product information
- Restock and promote high-demand products

Long term | 6–12 months

- Redesign for better garment accessibility
- Standardize sizing across styles
- Strengthen fabric and product quality controls

Business value

COT2 converts customer feedback into actionable product and experience improvements — helping the retailer respond faster, reduce avoidable returns, improve satisfaction, and support sales.

Capstone Recommendation: COT2 (Chain of Thought)

RECOMMENDED CAPSTONE BASELINE

COT2

Chain-of-Thought prompting

- Manual Text Macro-F1 : **79.73%**
- composite quality **:62.46%**
- Cost **: $0.125 / 1K**

What the evidence supports

- FS2 performed strongly on the initial seven-strategy benchmark.
- COT2 showed more consistent performance on the larger, evenly stratified evaluation.
- COT2 was lower-cost in the current run; FS2 was slightly faster.

Validation gate

Human-validate and test on a larger representative sample → establish monitoring, governance and cost controls before deployment.
Uncertainty check: Accuracy 90.29% (Wilson 95% CI 84.99–93.85%); Macro-F1 79.73% (2,000-review bootstrap 95% CI 70.81–87.41%; seed 42).

Task-specific quality weights are design choices, not universal standards. Rating alignment is secondary; cost and latency are separate from quality.

APPENDIX 01 | HOW MACRO-F1 IS CALCULATED

Macro-F1: Evaluation Flow

Use the same benchmark reviews, compare model predictions with the reference sentiment labels, calculate F1 for each sentiment class, then average the three class-level F1 scores.

1 — 50 benchmark reviews

Reference labels are established from the written review text.

2 — Run the model

Record the model's predicted sentiment for the same 50 reviews.

3 — Calculate 3 F1 scores

Calculate F1 separately for Positive, Negative and Neutral.

4 — Average equally

Give each sentiment class 33.3% weight and average the 3 F1 scores.

Example from the 50-review benchmark

Positive F1	Negative F1	Neutral F1
87.50%	92.06%	80.00%

Macro-F1 = (Positive F1 + Negative F1 + Neutral F1) ÷ 3

(87.50 + 92.06 + 80.00) ÷ 3

= 86.52%

KEY IDEA • The 50 reviews are not individually weighted. We calculate one F1 score for each sentiment class, then give the three classes equal importance.

LLM-as-a-Judge: one score built from five dimensions

The Judge evaluates the review + model output only. It does not see Rating or the reference sentiment label.

5% Sentiment Correctness

Did the output capture the overall customer experience?

20% Mixed / Nuanced Handling

Did it correctly resolve praise + complaint into the dominant sentiment?

10% Business Usefulness

Is the output practical for retail teams?

10% Factual Grounding

Are summary and insights supported by the review?

10% Output Completeness

Are all six requested fields present and usable?

Judge Score = 0.50×Sentiment + 0.20×Nuance + 0.10×Grounding + 0.10×Completeness + 0.10×Business Usefulness

Final Score = 55% Manual Text Macro-F1 + 20% Judge Score + 15% Mixed/Nuanced Handling + 10% Business Usefulness

Mixed-review example — qualitative evidence, not a separate benchmark score

"I love the wide straps but it runs small and is short."
This illustrates the mixed-signal problem: praise for design, but meaningful fit/length concerns. The qualitative analysis found COT approaches showed slightly better sensitivity to mixed feedback, although this type of case was not handled consistently enough to make it a standalone selection criterion.